

LESSON 3-7 • SESSION 2 • PRACTICE • FORM A

Multi-Step System Conversions and Dosage

Learning target – I can solve a multi-step problem that requires converting units before using a rate. · 6.RP.A.3d

REMEMBER

- Start with the amount you are given. Change it to the unit the **next step** needs. Then use the **rate**.
- Write the **unit** after every answer: lb, then kg, then mg, then mL.
- For area, multiply length $\times$ width, then divide by what one bag or can covers. Round **up** for whole items.

WORKED EXAMPLE

A cat weighs 11 lb. The dose is 3 mg per kg. The syrup has 5 mg per mL. (1 kg $\approx$ 2.2 lb)

Step 1 $11 \div 2.2 = 5$ kg.

Step 2 $5 \text{ kg} \times 3 \text{ mg per kg} = 15 \text{ mg}$.

Step 3 $15 \text{ mg} \div 5 \text{ mg per mL} = 3 \text{ mL}$.

Answer 3 mL of syrup

WORDS multi-step (*de varios pasos*) · dose (*dosis*) · per (*por*)

PRACTICE*show your work · label units*

1. **Max the Dog.** Complete each step. (1 kg $\approx$ 2.2 lb)

a. 33 lb $\approx$ _____ kg

b. 15 kg $\times$ 5 mg per kg = _____ mg

c. 75 mg $\div$ 15 mg per mL = _____ mL

d. 25 kg $\approx$ _____ lb

2. **Pony Medicine.** A pony weighs 187 lb. The medicine dose is 2 mg per kg of body weight. (1 kg $\approx$ 2.2 lb)

Hint Change pounds to kilograms before you use the dose.

a. Weight in kilograms: _____ kg

b. Total dose: _____ mg

Show your work.

3. **Elephant Calf.** A baby elephant weighs 110 kg. (1 kg $\approx$ 2.2 lb, 1 ton = 2,000 lb)

Hint Multiply by 2.2, then subtract from 2,000.

a. Weight in pounds: _____ lb

b. Pounds more to reach 1 ton: _____ lb

Show your work.

4. **Gas Tank.** A car's tank holds 57 L. Gas costs \$3.50 per gallon. (1 gal $\approx$ 3.8 L)

a. Tank in gallons: _____ gal

b. Cost to fill: _____

Show your work.

5. **Butter.** A recipe calls for 280 grams of butter. About how many ounces is that? (1 oz $\approx$ 28 g)

Hint An ounce is heavier than a gram, so divide.

Show your work.

Answer _____ oz

6. **Dog Medicine.** A dog weighs 66 lb. The dose is 4 mg per kg. Find the dose. (1 kg $\approx$ 2.2 lb)

Hint Kilograms first, then multiply by 4 mg.

Show your work.

Answer _____ mg

7. **Travel Cost.** A trip is 240 km. A car uses 1 L of fuel every 12 km. Fuel costs \$1.80 per liter.

Hint Divide the distance by 12 km per liter.

a. Liters used: _____ L

b. Fuel cost: _____

Show your work.

8. **Class Garden.** A garden bed is 6 ft by 9 ft. One bag of soil covers 6 square feet and costs \$4.25.

a. Area of the bed: _____ ft²

b. Bags needed: _____ bags

c. Total cost: _____

d. What changes if one bag covers 9 square feet? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Marcus's Work.** Marcus converted a 44-lb dog to kilograms by multiplying: $44 \times 2.2 = 96.8$ kg. Explain why 96.8 kg is impossible. Then find the correct weight. (1 kg $\approx$ 2.2 lb)

TEST PRACTICE

choose the best answer

10. A 44-pound dog is about how many kilograms? (1 kg $\approx$ 2.2 lb)

A 96.8

B 46.2

C 41.8

D 20

11. At 5 mg per kg, a 12-kg animal needs –

A 2.4 mg

B 60 mg

C 17 mg

D 7 mg

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

170 O	\$38.25 S	\$199.50 X	\$52.50 A	5 D	1,758 S	120 E	10 G	264 Z	374 Q
#1c		#2b	#3b	#4b	#5	#6	#8c		

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____